A formally verified continuous nonlinear Matkowski solution

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Abstract

We give a complete Lean certificate for a concrete continuous nonlinear solution of the quasi graph-additive functional equation. This is an explicit instance of the counterexample mechanism already published by Kiss, not a new disproof. The MathDB record still labels the question open.

Theorem 1. *The continuous function* $f(x) = \min(\max(x, 0), 1)$ *satisfies*

$$f(f(-x) + x) = f(-f(x)) + f(x) \qquad (x \in \mathbb{R}),$$

but there are no $a, b \in \mathbb{R}$ *such that* $f(x) = ax$ *for* $x \le 0$ *and* $f(x) = bx$ *for* $x > 0$.

Proof. Minimum and maximum preserve continuity of real functions, so f is continuous. It is zero on the nonpositive half-line, nonnegative everywhere, and satisfies $f(t) \le t$ for $t \ge 0$. Thus $f(-f(x)) = 0$. For $x \ge 0$ we have $f(-x) = 0$, so the asserted identity reduces to $f(x) = f(x)$. For $x < 0$, the bound $f(-x) \le -x$ implies $f(-x) + x \le 0$, so both sides are zero. This proves the identity everywhere.

If the asserted two-slope representation existed, the values $f(1) = f(2) = 1$ would give $b = 1$ and $2b = 1$, a contradiction. □

Attribution. Kiss [1] already disproves Matkowski's proposed classification. His Corollary 1 permits an arbitrary positive-half-line part between zero and the identity, with zero slope on the negative half-line. Our f is an elementary instance. The present result is a complete formal verification and explicit example of that known mechanism. It remains a counterexample if the conjecture's extra restrictions on a, b are dropped.

Formal coverage and source status. The main declaration is `MatkowskiCounterexample.continuous_nonlinear_counterexample` in `formal/Matkowski.lean`. All real inputs, continuity, and failure of the whole proposed two-slope class are covered. The build checks every local declaration against a pinned Lean/mathlib environment and permits only standard foundational axioms.

The exact MathDB 373182 record had no posted solutions and status `open` at retrieval. Its own cited primary source [1] explicitly establishes a disproof; hence this should be treated as a formalization of a known resolution and a source-status correction, not as a newly solved open problem.

References

[1] T. Kiss, *A Counterexample to Matkowski's Conjecture for Quasi Graph-Additive Functions*, arXiv:2602.15548v1, equation (1), Proposition 1 and Corollary 1. https://arxiv.org/html/2602.15548v1.